

EP 321: Foundations of Electrodynamics

Course Code: EP 321

Course Name: Foundations of Electrodynamics

L-T-P-C: 2.5-0.5-0-3

Prerequisites: None (Discipline Core)

Intended for: B.Tech Engineering Physics

Mutual Exclusion: None

Approval: 50th BoA (B.Tech Engineering Physics curriculum, 3-0-0-3); revised in 60th BoA (formerly IC 221 Foundations of Electrodynamics, recoded as EP 321, L-T-P-C 2.5-0.5-0-3)

Preamble

This course offers a unified introduction to classical electrodynamics, starting from the fundamentals of electrostatics and magnetostatics to the full formulation of Maxwell's equations. Part I develops the mathematical and physical foundations, including vector calculus, boundary conditions, dielectric and magnetic media, and electromagnetic induction. Part II applies Maxwell's equations to wave propagation in vacuum and matter, gauge invariance, dispersion, and energy-momentum conservation, and explores modern topics like negative-index materials. The course concludes with guided-wave theory, radiation, and antennas, equipping students with the tools to analyze electromagnetic phenomena in both free space and materials.

Course Modules with Quantitative Lecture Hours

Part I – Derivation of Maxwell's Equations

- **Review of Vector Calculus and Electrostatics:** Vector calculus, Helmholtz equation, Coulomb's law, Gauss's law, Poisson and Laplace equations. (6 Lectures)
- **Electrostatic Boundary Conditions and Dielectrics:** Conductors and capacitors; mean-value and uniqueness theorems; separation of variables; dipoles and electric polarization in matter (dielectrics). (6 Lectures)
- **Magnetostatics and Magnetic Materials:** Lorentz force law; Biot-Savart law and magnetic vector potential; boundary conditions on $\mathbf{B}$; magnetic materials (paramagnetic, diamagnetic); bound currents; boundary conditions on $\mathbf{H}$; inductance and magnetic energy density. (6 Lectures)
- **Electrodynamics and Maxwell's Equations:** Ohm's law; electromotive forces; Faraday's law; full Maxwell's equations in differential and integral form. (5 Lectures)

Part II – Maxwell's Equations and Electromagnetic Waves

- **Waves in Vacuum and Energy-Momentum Conservation:** Electromagnetic waves in vacuum; Maxwell's stress tensor; momentum conservation; Poynting theorem (energy and momentum conservation). (5 Lectures)
- **Gauge Transformations:** Gauge freedom; Coulomb gauge and Lorentz gauge choices. (3 Lectures)
- **Waves in Matter and Dispersion:** Reflection, transmission and polarization at interfaces; electromagnetic waves in dispersive media; Kramers-Kronig relations; Lorentz oscillator model for dispersion and absorption; negative-index materials. (6 Lectures)



- **Waveguides and Radiation:** Rectangular and circular waveguides: TE and TM modes; radiated power; electric dipole radiation; basic antenna theory. (6 Lectures)

Text Book

1. D. J. Griffiths, *Introduction to Electrodynamics*.

References

1. R. P. Feynman, *Lectures on Physics Vol. II*.
2. D. K. Cheng, *Fields and Waves in Electromagnetics*.
3. M. N. O. Sadiku, *Elements of Electromagnetics*.
4. E. M. Purcell and D. J. Morin, *Electricity and Magnetism*.
5. B. B. Laud, *Electromagnetics*.
6. J. D. Jackson, *Classical Electrodynamics*.